31N0402

Roll No

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B.Tech III Sem Main/Back Exam 2025-26

Computer Science and Engineering

(3CS4-02) Digital Electronics

3AD4-02, 3AM4-02, 3AI4-02, 3CM4-02, 3CY4-02, 3CD4-02, 3CS4-02, 3IT4-02, 3IC4-02 - Common to CS, IT, AI, DS,MC, CM, CD, CA, AD, AM, CY, IO

Time : 3 Hours

Maximum Marks: 70
Min. Passing Marks:

Instructions to Candidates:

Part-A: Short answer questions (up to 25 words) 10x2 marks = 20 marks. All ten questions are compulsory.

Part- B: Analytical/Problem Solving questions 5x4 marks = 20 marks. Candidates have to answer 5 questions out of 7.

Part – C: Descriptive/Analytical/Problem Solving questions 3x10 marks = 30 marks. Candidates have to answer 3 questions out of 5.

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1__NIL_____ 2_____NIL_____

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Part-A

[2x10=20]

Q.1. Convert the following binary numbers to decimal: [2]

a. 110101

b. 101001

Q.2. Convert A+B to minterms. [2]

Q.3. What are the three basic parts of a BCD adder circuit? [2]

Q.4. Give four basic flip-flop applications. [2]

Q.5. What does the term logic family signify? [2]

Q.6. What are the limitations of fixed-point numbers? [2]

Q.7. What is the significance of DeMorgan's theorem? [2]

Q.8. Draw the block diagram of a half-subtractor. [2]

Q.9. What is meant by the term edge-triggered? [2]

Q.10. When and where do we use open-collector operation? [2]

PART-B

[5x4=20]

Q.1. Illustrate and describe the format for single precision binary floating- point numbers with the help of an example. [4]

Q.2. Simplify the following equation using the Karnaugh map procedure: [4]

$$\times = \overline{ABC} + A\overline{CD} + A\overline{B} + ABC\overline{D} + \overline{ABC}$$

Q.3. Implement a full-adder using Karnaugh maps. [4]

Q.4. Explain in detail the working of Finite state machines. [4]

Q.5. Design a MOD-5 up-counter that counts in the sequence 6-7-8-9-10-6-7-8-9-10-, and so on. [4]

Q.6. What is the difference between Boolean algebra and ordinary algebra? Draw the combination of AND, OR, and NAND gates to provide the XOR function. [4]

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Q.7. Illustrate and explain the operation of CMOS-NOR gate. [4]

PART- C

[3x10=30]

Q.1. Draw the logic circuit for the following equations. Apply DeMorgan's theorem and Boolean algebra rules to reduce them to equations having inversion bars over single variables only. Draw the simplified circuit: [10]

a. $W = \overline{AB} + \overline{A+C}$

b. $X = A\overline{B+C} + \overline{BC}$

c. $Y = \overline{(AB)+C} + B\overline{C}$

d. $Z = \overline{AB + (\overline{A}+C)}$

Q.2. Use Karnaugh map to simplify the following equations: [10]

a. $E = \overline{A}BC + \overline{B}CD + AC + \overline{ABC}\,\overline{D}$

b. $E = ABC + BCD + AC + BC$

Q.3. Discuss the detailed procedure for the design of combinational circuits. [10]

Q.4. a. Illustrate and explain the conversion of S-R flip-flop into a J-K flip-flop. [10]

b. Compare combinational and sequential circuits.

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Q.5. With a circuit diagram explain the operation of the CMOS NAND gate. What are its advantages over corresponding TTL gates? [10]

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